ADITYA BISHT

Agentic AI Engineer | Multi-Agent Systems | RAG & Tool Calling | LLM Evaluation

+91 7982759592 | New Delhi | aaditya85677@gmail.com | LinkedIn | GitHub | LeetCode | CodeChef

SUMMARY

Final-year B.Tech IT student (CGPA 8.60) who builds production-grade AI agents and the evals that prove they work. Shipped 4 open-source frameworks (multi-agent orchestration, 8-strategy RAG, LoRA fine-tuning, LLM security), each live on HuggingFace with measured accuracy, latency and cost tradeoffs. At Synngular AI Labs, built multi-agent workflows with tool calling and human-in-the-loop gates, and deployed self-hosted LLMs behind a secured inference API. Treats LLM unreliability as a design constraint to evaluate and engineer around.

EXPERIENCE

Prompt Engineering Intern — Synngular AI Labs Jun 2026 – Sep 2026 | Remote

- Built **Project Atlas**, a 4-agent workflow (inventory watch, PR approval, vendor scoring, RFQ generation) with per-plant scoping and external tool calls (MailGun), plus a **10-agent LangGraph** planning system with human-in-the-loop approval gates.
- Designed a Slack-integrated multi-agent system (Task Logger, Health Monitor, Meeting agents); benchmarked it against a single-agent design and shipped the architecture that held up on real workflows.
- Benchmarked **3 agentic architectures** (Single Agent, Planner+Implementor, Multi-Agent+Router) on a SQLite REST API spec using a 6-criterion weighted rubric (security, correctness, quality, spec compliance, robustness, efficiency). The Router passed 16/16 surface tests yet scored lowest (~52%), exposing that platform-verifier scores were unreliable.
- Deployed **Qwen3.6-27B-FP8** and **Qwen3.8-27B-FP8** on a DGX Spark (GB10) with vLLM: per-model tool-call parsers, FP8 quantization and thinking mode off to speed up Cline, the team's coding agent. Secured the endpoint with a FastAPI gateway (API-key auth, Redis rate limiting), Caddy TLS and Cloudflare.
- Built and tested CLI-based agent skills (SKILL.md, tool.json, strict skill-grounded execution) across docx, PDF and Next.js UI generation; documented gaps between platform docs and actual behavior.

PROJECTS

AgentBench — Single vs Multi-Agent LLM Benchmarking Framework GitHub | Live Demo

Python | LangGraph | Groq | FAISS | Tavily | SQLite | Streamlit | Plotly

- Designed a 5-node pipeline (Planner, Researcher, Analyst, Writer, Memory) with Tavily tool calls, Pydantic structured outputs and SQLite agent memory; benchmarked 50 queries across 5 metrics and 2 agent configurations.
- Built a resumable LLM-as-judge runner scoring 100 responses on accuracy, hallucination, relevance, latency and cost: multi-agent cut hallucination **6× (24% → 4%)** at a **56× latency cost** (307s vs 5.5s).
- Shipped a Streamlit dashboard with 7 live Plotly charts and 8 metric cards for real-time comparison.

RAG Systems Eval Suite — 8-Strategy RAG Benchmark Framework GitHub | Live Demo

Python | FAISS | BM25 | LangChain | Groq | Cerebras | Streamlit | Pydantic

- Benchmarked 8 strategies (Naive, Hybrid, Reranking, HyDE, Query Rewriting, Advanced, Adaptive, Base LLM) over 4,000 answer-metric pairs (50 questions × 8 systems × 10 metrics).
- FAISS + BM25/RRF hybrid search with cross-encoder reranking lifted faithfulness **16% (0.760 → 0.881)** and cut hallucination **4× (0.816 → 0.221)**.
- Designed a dual-provider LLM-as-judge (Groq Llama-3.3-70B + Cerebras Llama-3.1-8B) with SQLite checkpoint/resume; published as an **arXiv paper (cs.IR)**.

RAG Security Defense — Multi-Layer LLM Guardrails & Hybrid Retrieval GitHub | Live Demo

Python | Streamlit | Groq | ChromaDB | BM25 | Cross-Encoder Reranking | RBAC

- Built a 5-layer defense pipeline that blocked **100% of 30 labeled attacks** (15 regex injection + 15 jailbreak) at <1ms Layer-1 latency, validated on a 100-query suite.
- Implemented 4-stage hybrid retrieval (dense + BM25 + RRF k=60 + cross-encoder, 40 candidates to top-5) with 3-tier RBAC and **0 cross-tier leaks**.

CodeSage — LLM vs RAG vs Fine-Tuning Comparative AI Framework GitHub | Live Demo

Python | LangChain | FAISS | HuggingFace PEFT | LoRA | Streamlit | Groq API

- Fine-tuned Qwen2.5-1.5B with LoRA/PEFT: **85.3% accuracy at $0.0002/query with 0% hallucination**, vs baseline LLM (43.2% hallucination) and RAG (0.768 answer relevance), quantifying the accuracy/cost/latency tradeoff.

SKILLS

Agentic AI: LangGraph, LangChain, CrewAI, MCP, Multi-Agent Systems, Tool/Function Calling, Planning & Reasoning, Agent Memory & State, Human-in-the-Loop, Agent Evaluation

LLM / RAG: OpenAI, Claude, Gemini, Groq, Cerebras, vLLM, Prompt Engineering, Structured Outputs, Embeddings, Hybrid Search (BM25/RRF), Reranking, LoRA/PEFT, LLM-as-Judge, HuggingFace, PyTorch, TensorFlow

Engineering: Python, FastAPI, REST APIs, Async, SQL, Redis, Docker, Git/GitHub, Linux, n8n

Data & Cloud: ChromaDB, FAISS, Pinecone, SQLite; AWS (S3, EC2, Lambda), OCI, GitHub Actions (CI/CD), MLOps

EDUCATION & CERTIFICATIONS

MAIT, GGSIPU — B.Tech Information Technology 2023 – 2027 | CGPA: 8.60

Kendriya Vidyalaya Janakpuri, CBSE Class 12: 88.2% | Class 10: 94.2%

Oracle Cloud Infrastructure 2024: Generative AI Professional | AI Vector Search Professional